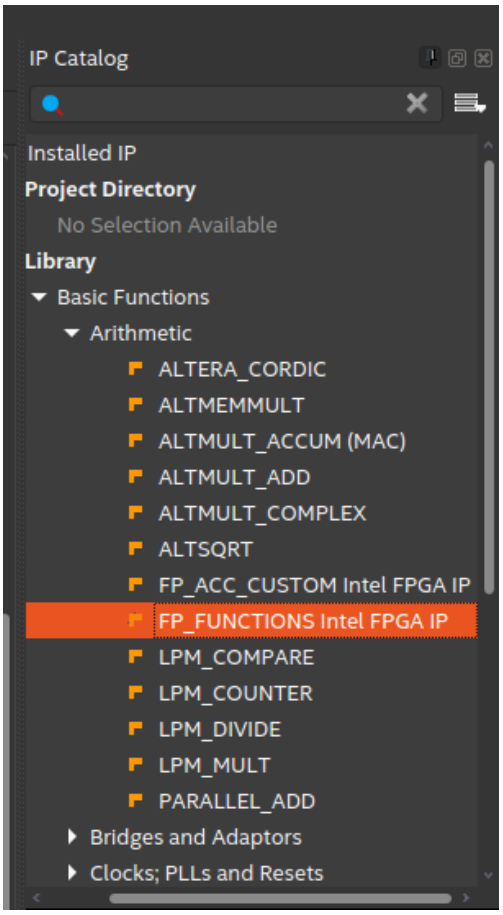
Herbert Wertheim College of Engineering
Department of Electrical & Computer Engineering
UNIVERSITY of FLORIDA

Newman Waters, PI
EEL4712C - Spring 2025

Lab 6 Additional Info

Part 1



Quartus IP Manager
You can access the Quartus IP Manager by going to **Tools > IP Catalog**
This will pull up on the right side of your screen. You can then follow the lab instructions step-by-step to generate the components.
Note: On Linux, you need to resize the windows that appear before doing anything, or else the IP manager will crash.

MIF Files
MIF Files are used to define the initial contents of memory blocks, such as RAM or ROM, in FPGA designs. It is a simple text file that specifies how memory should be preloaded when the FPGA is programmed.

A MIF file consists of two main sections:

1. **Header Section** – Defines memory properties (depth, width, etc.).
2. **Content Section** – Lists memory addresses and their corresponding data values.

```
DEPTH = 16;            -- Number of memory locations
WIDTH = 8;             -- Bits per memory word
ADDRESS_RADIX = DEC;   -- Address format (DEC, HEX, BIN)
DATA_RADIX = HEX;      -- Data format (DEC, HEX, BIN)
CONTENT BEGIN
  0  :  2A;           -- Address 0 contains 0x2A
  1  :  FF;           -- Address 1 contains 0xFF
  [2..5] :  00;       -- Addresses 2 to 5 contain 0x00
  6  :  1C;
  7  :  3E;
END;
```

Example MIF File

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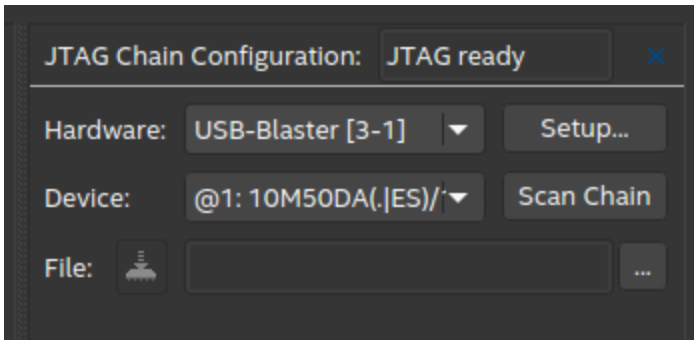
Part 2

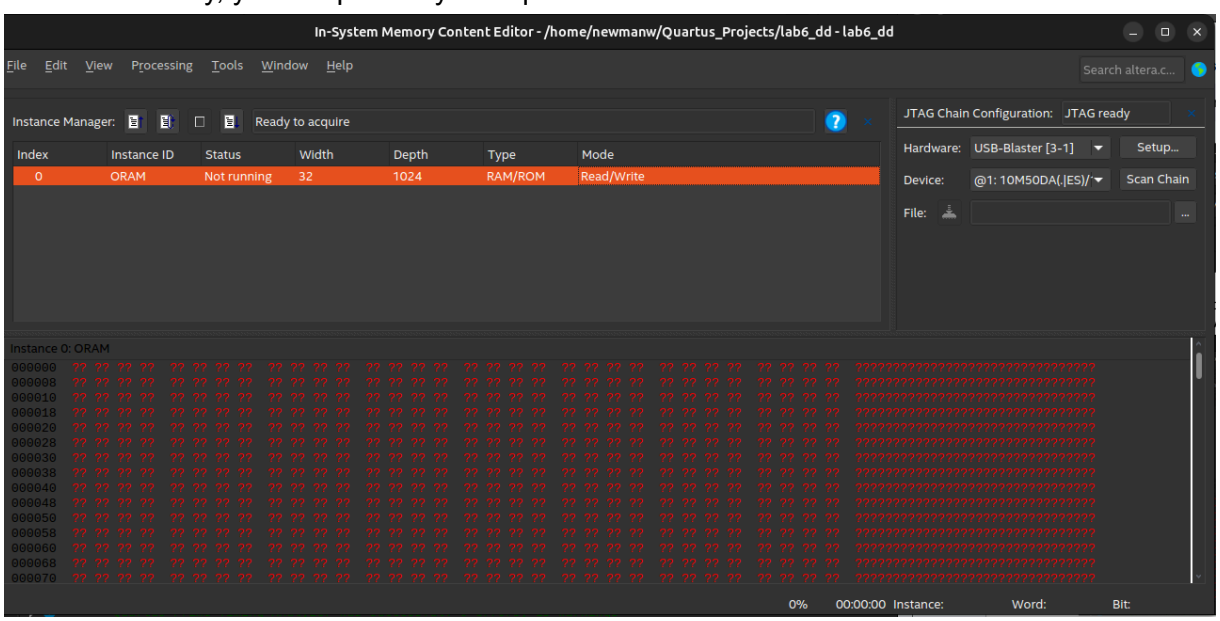
Using the In-System Memory Content Editor

To modify and read the memory contents of the RAM module you will make for part 2, you will navigate to **Tools > In-System Memory Content Editor**

Make sure that you select your device on the side, similar to how you would do it when programming your board:



You have multiple options when using the In-System Memory Content Editor. When you open the editor initially, you will probably see question marks:

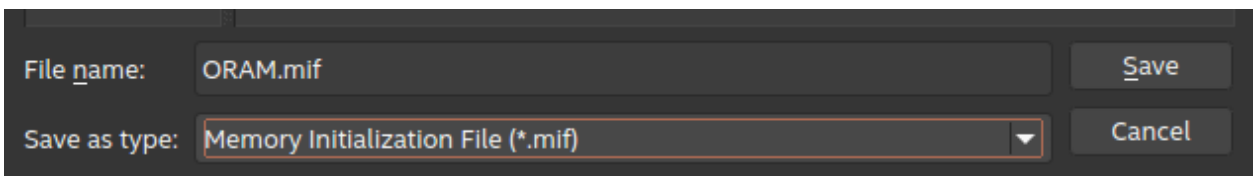


Click on the RAM instance (highlighted orange).
Then, navigate to **Processing > Read Data from In-System Memory.**

This will appear as a bunch of zeros until your board has completed its processing.

So, start the processing (flip the start switch on and off), and wait until the READY signal is asserted (assigned LED turns on). Then, navigate back to **Processing > Read Data from In-System Memory.**

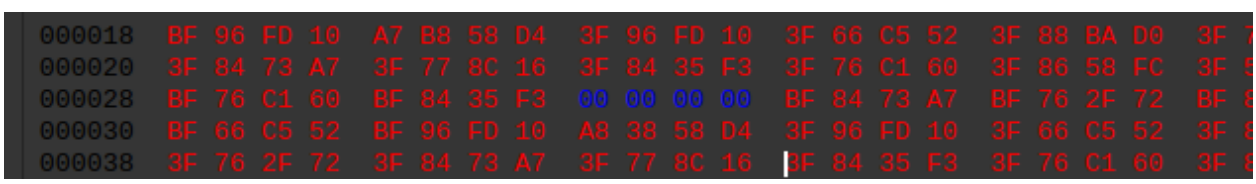
You should now see your data appear in place of the question marks. Once the data has appeared, go to **Edit > Export Data to File** and save it as a .MIF file (it will probably default to try and save as a .HEX file, so make sure to change it as highlighted below).



Save it somewhere that is easily accessible to you, and include it in your submission.

Additional Notes (Writing to the RAM):

You can edit the MIF file manually by clicking on any byte in the memory. The word (4 bytes) will be highlighted blue with any changes you make, and will remain blue:



To actually write these changes, you can go to **Processing > Write Data to In-System Memory**. This actually makes the changes on the RAM, and the blue text will disappear: